

DANIAL AMIN

AI & Data Science Lead | Staff AI Engineer | AI Research Scientist

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer specializing in production-grade LLM systems, agentic AI frameworks, and scalable ML infrastructure. Deep technical expertise in architecting and deploying RAG systems, vector databases, and MLOps pipelines from prototype to production. Proven track record building enterprise-scale AI applications using LangChain, LangGraph, PyTorch, and TensorFlow across Azure, GCP, and AWS platforms. Strong foundation in deep learning, reinforcement learning, computer vision, and NLP with hands-on experience optimizing models for performance, latency, and cost efficiency. Currently advancing state-of-the-art AI systems at Samsung Design Innovation Center while pursuing doctoral research in AI at University of Vaasa.

CORE COMPETENCIES

AI Frameworks & Tools: LangChain, LangGraph, LangServe, AutoGen, CrewAI, PyTorch, TensorFlow, Scikit-learn, Hugging Face
LLM & GenAI: RAG Systems, Prompt Engineering, Fine-tuning, LLM Evaluation, Agentic AI, Multi-modal AI, Model Optimization
ML Engineering: Production ML Pipelines, Model Deployment, A/B Testing, Feature Engineering, Model Monitoring, GPU Optimization
Data & Vector Technologies: Pinecone, Weaviate, ChromaDB, FAISS, Elasticsearch, ETL/ELT Pipelines, BigQuery
Deep Learning: CNNs, RNNs, Transformers, GANs, Reinforcement Learning, Computer Vision, NLP, Time Series
Programming & Libraries: Python, SQL, PySpark, Pandas, NumPy, CUDA, PyTorch, TensorFlow, Keras
DevOps & MLOps: Docker, Kubernetes, CI/CD, Git, MLflow, DVC, Azure DevOps, Airflow
Cloud & Infrastructure: Azure (Azure ML, AKS, OpenAI), GCP (Vertex AI, BigQuery), AWS (SageMaker, Lambda)
Technical Architecture: System Design, API Development (REST/GraphQL), Microservices, Data Pipeline Design
Languages: English (C1), Urdu (C2), Punjabi (C1)

EXPERIENCE

- Senior AI & DS Consultant** May 2026 - Present
Illusian Tech Office (Impact Lab)
 - Leading the generative AI evaluation framework to assess impact of generative AI on society.
- Senior AI & DS Consultant** Nov. 2024 - Apr. 2026
INTM Groupe
 - Client: ENI Plenitude**
 - Leading the data science practice, developing solutions for the marketing, energy consumption, prediction and forecasting.
 - Creating dedicated, orchestrated solutions for churn prediction due to price sensitivity.
 - Developed a simulator to predict the effect of price on churn, acquisition, and expected margins.
 - Developing a emotional AI analyst, capable of identifying patterns in textual communication with clients.
 - Leading a team of four early to mid level practitioners and supporting them in their learning of better applied practices of data science especially in production.
 - Client: Samsung Design Innovation Center (SDIC)**
 - Led end-to-end delivery of AI Design Companion (AIDC) project, managing client stakeholder relationships and translating design team requirements into technical solutions combining generative and predictive AI models.
 - Architected multi-stage unstructured database serving as ground truth for AI applications, designing data ingestion pipelines and ETL workflows to support design optimization use cases.
 - Developed production APIs using FastAPI to integrate AI capabilities into client's existing design workflows, ensuring seamless adoption and minimal disruption to current processes.
 - Provided technical support for AI Research Companion (AIRC) project, implementing RAG architecture with vector databases for semantic search across design documentation, achieving 40% improvement in retrieval accuracy.
 - Established MLOps practices including automated testing, model versioning, and continuous deployment pipelines aligned with client's infrastructure requirements.
- AI & Data Science Lead** Sep. 2023 - Nov. 2024
i13 Ventures Studio
 - Provided AI/ML consulting services to 6+ startup clients, delivering enterprise RAG solutions using LangChain, Pinecone, and Azure OpenAI, implementing hybrid search strategies that reduced query latency by 60%.
 - Built production agentic AI systems using LangChain and LangGraph for document processing, data extraction, and workflow automation, incorporating enterprise-grade error handling and monitoring.
 - Established MLOps infrastructure including CI/CD pipelines, automated monitoring, and A/B testing frameworks, reducing model deployment cycles from weeks to hours.
 - Designed scalable data pipelines processing 10TB+ monthly volume across multi-modal data (text, image, audio) using Apache Airflow and cloud storage solutions.
 - Developed REST and GraphQL APIs for ML model serving with comprehensive security, versioning, and rate limiting for production environments.
- Head of Data Analytics — Lead Data Scientist** Feb. 2023 - Sep. 2023
Abacus Global
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- ▶ Led development of consulting practice and technical team, establishing service delivery frameworks and best practices for enterprise client engagements.
 - ▶ Delivered end-to-end ML solutions for enterprise clients across multiple industries, establishing DevOps practices including containerization, orchestration, and automated deployment pipelines.
 - ▶ Designed big data processing infrastructure on GCP using BigQuery, Dataflow, and Pub/Sub, implementing stream and batch workflows handling 5M+ daily events.
 - ▶ Built RESTful APIs for predictive model deployment with load balancing, caching, and monitoring dashboards for production systems.
- **Technical Consultant - Data Science** Feb. 2022 - Feb. 2023
Automat Solutions Inc via Turing Remote, USA
- ▶ Engaged as technical consultant to deliver smart battery optimization solution using deep reinforcement learning, implementing GPU-accelerated training pipelines and deploying models for edge inference with TensorRT.
 - ▶ Built automated data pipelines processing 100K+ sensor data points per second with real-time anomaly detection capabilities.
 - ▶ Developed Flask-based REST APIs for model inference with request queuing, caching, and monitoring achieving 99.9% uptime.
- **ML Consultant** Feb. 2021 - Feb. 2022
Omdena Inc Remote, USA
- ▶ Provided ML consulting services to two NGO/NPO clients, delivering computer vision solutions implementing transfer learning and model optimization techniques, achieving 12% accuracy improvement over baseline.
 - ▶ Built automated ML pipelines using Python and Docker for model training, evaluation, and deployment with version control and CI practices.
 - ▶ Developed FastAPI endpoints for model serving with batch prediction and real-time inference capabilities (sub-100ms latency).
- **Assistant Manager - Aerospace Analytics** Aug. 2017 - Nov. 2022
Government of Pakistan On-site, Pakistan
- ▶ Delivered predictive maintenance ML system for aerospace applications, building data integration pipelines and automated alert systems reducing maintenance costs by 35%.
 - ▶ Developed fuel consumption optimization models with comprehensive data preprocessing, feature engineering, and automated retraining schedules.

EDUCATION

- **University of Vaasa** Jan. 2025 – Date
Doctorate of Science Hybrid, Finland
- **Rennes School of Business** Sep. 2023 – Sep. 2024
Masters in Business Administration On-Campus, France
- **MIT Institute for Data, Systems and Society** Jan. 2024 – May 2024
Data Science and Machine Learning: Making Data-Driven Decisions Distance Learning, USA
- **University of Wisconsin, Madison** Aug. 2022 – May 2023
Micro Masters in Business: Data, Insights and Analytics Distance Learning, USA
- **University of California San Diego** Aug. 2021 – Mar. 2023
Micro Masters in Data Science Distance Learning, USA
- **Massachusetts Institute of Technology** Aug. 2019 – Nov. 2023
Micro Masters in Statistics and Data Science Distance Learning, USA
- **Institute of Space Technology** Sep. 2013 – Aug. 2017
B.S in Aerospace Engineering On-Campus, Pakistan

KEY PROJECT DELIVERIES

- **AIDC - AI Design Companion (Samsung SDIC)**
 - ▶ Primary project lead delivering comprehensive AI design tool combining generative and predictive AI models for data-driven design optimization at Samsung Design Innovation Center.
 - ▶ Architected multi-stage unstructured database infrastructure serving as ground truth for AI models, implementing data pipelines for design data ingestion, processing, and retrieval workflows.
 - ▶ Built agentic AI system using LangChain for autonomous design assistance, implementing multi-agent workflows for design analysis and optimization recommendations.

🔗 **Tech Stack:** Python, LangChain, LangServe, LangGraph, PyTorch, Directus, FastAPI, Docker
- **AIRC - AI Research Companion (Samsung SDIC)**
 - ▶ Supported delivery of RAG system for Samsung Design Innovation Center, implementing semantic search across design documentation and research materials.
 - ▶ Built hybrid data pipeline processing unstructured design and research data, implementing ETL workflows for document ingestion, chunking, embedding generation, and vector storage.

🔗 **Tech Stack:** Python, LangChain, FAISS, OLLAMA, FastAPI, Docker, CI/CD

- **Bob - The Startup Advisor (i13 Ventures)**

- ▶ Delivered enterprise RAG system using LangChain and Pinecone, implementing multi-stage retrieval pipeline with query optimization and reranking for startup evaluation across multiple innovation hubs.
- ▶ Built scalable API infrastructure using FastAPI and LangServe with load balancing, rate limiting, and authentication, serving 500+ requests/hour.

🔗 **Tech Stack:** Python, LangChain, LangServe, Pinecone, Azure OpenAI, FastAPI, Docker, Azure DevOps

- **Sandy - The Financial Advisor (i13 Ventures)**

- ▶ Delivered agentic AI system using LangChain and LangGraph for autonomous financial analysis, implementing multi-agent architecture with specialized agents for data retrieval, analysis, and report generation.
- ▶ Built data ingestion pipeline processing financial statements and market data, implementing ETL workflows with validation, transformation, and vector database storage.

🔗 **Tech Stack:** Python, LangChain, LangGraph, Pinecone, Azure, FastAPI, Celery, Redis

- **Integrated Intelligence Platform (Abacus Global - Caribbean Banking Client)**

- ▶ Delivered comprehensive data pipeline architecture on GCP, implementing ETL workflows using Dataflow, BigQuery for data warehousing, and Looker for visualization across three departments.
- ▶ Built automated data ingestion from SAP S4 HANA, implementing data quality checks, transformation logic, and incremental loading strategies for real-time insights.

🔗 **Tech Stack:** GCP, BigQuery, Dataflow, Looker, Python, Cloud Run, SAP S4 HANA, Docker

🌟 CERTIFICATIONS

● Certified AI Practitioner by AWS	AWS
● Introduction & Advanced Topics of Model Context Protocol by Anthropic	Anthropic
● Expert Certification in Artificial Intelligence by European Information Technologies Certification Institute	EITCA
● Professional Certificate in Data Analytics by Google	Coursera
● Professional Certificate in Project Management by Google	Coursera
● Professional Certificate in Data Science by IBM	Coursera
● Specialization in Deep Learning by Deeplearning.ai	Coursera
● Specialization in Natural Language Processing by Deeplearning.ai	Coursera
● Complete Career track of Data Scientist by 365DataScience	365DataScience
● Expert track in Data Science Fundamentals on Microsoft Azure by CloudSwyft and Microsoft	FutureLearn
● Nano degree in AI Programming with Python by Udacity	Udacity
● Developer Certification in Data Analysis with Python by FreeCodeCamp	FreeCodeCamp